An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

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1. Executive summary

Floods displace millions of people in India each year. The limiting factor in a rescue is not reaching survivors but locating them.

We propose three one-metre, 6.4 kg multirotor aircraft that search flooded ground as a coordinated group under one operator, identify people from the air, report their position to about a metre, and deliver a relief kit to each — without mobile network or internet.

Sizing, error budgets and flight software are complete and reproducible; coordination has been demonstrated across three simulated autopilots. No aircraft has yet flown, so the figures here are calculated, and the programme is ordered to replace them with measurements.

The request is **INR 7,43,004 across 32 phases**, none above INR 30,000, each released on the result of the preceding one.

2. The problem

After a flood the binding constraint is time-to-locate. Boat teams search slowly, at risk, and cannot see rooftops. Helicopters are scarce and committed to evacuation. Conventional drones need one pilot each, depend on infrastructure the disaster has removed, and cannot assist a survivor once found.

Detection is the harder problem. A person in floodwater presents head and shoulders only, about 40 cm across. Detection models take a fixed, small input, so the usual approach shrinks the image and the target shrinks with it, to a size at which published accuracy is 29–61 %. A smaller, faster model is worse, because smaller models take smaller inputs.

Flooding adds two constraints: no dataset annotates *people* in flood imagery, and thermal imaging fails in water, where skin equilibrates toward water temperature.

3. Our solution

System. The operator draws a search region of any shape; the software splits it into three balanced sub-regions and plans a path over each — verified on thirteen non-convex shapes, no path leaving the region, imbalance below 0.02 %. The aircraft then fly autonomously, and the test harness demonstrates structurally that no command is sent after the mission begins. Detection and geolocation run on board. Each aircraft carries four kits and can descend, null its groundspeed over a survivor, and release one. Link loss, low battery and operator abort converge on a single return-and-land behaviour.

Detection. The image is not shrunk. Each frame is divided into 48 overlapping tiles processed at native resolution, and the survey altitude is 40 m rather than 60 m. Together these place the target at **39 pixels** rather than 13 — nine times the area, and outside the band where detector performance collapses. Both changes are needed; either alone leaves the target inside it.

This costs four times the computation and was verified feasible first: 48 tiles at 2 Hz is 96 inferences per second against 130–160 measured for the accelerator. Mission duration is marginally shorter than at 60 m, the lower climb outweighing the extra transects.

One item is open. Confirming a survivor across twelve frames needs about 3 Hz, which exceeds that capacity. A two-stage filter — a low-cost model discarding the roughly 87 % of tiles that are open water — would recover it. It will be evaluated against public data first, since such a filter fails without indication when it discards a tile containing a person.

4. Deliverables

1. Three aircraft operating as a coordinated group under one operator.
2. A ground station that displays mission state and is architecturally incapable of commanding the aircraft.
3. Autonomy software — partitioning, path planning, allocation, failsafes — with an automated test suite.
4. Measured detection accuracy and a verdict on the two-stage filter, replacing the calculated figures.
5. A publication: tiling, aerial person detection and low-power onboard inference have not been combined for flood conditions.
6. Source code and results published in either outcome.

5. Benefits

Operational. Ten hectares surveyed in under eight minutes, returning a list of coordinates rather than a video feed needing continuous attention.

Economic. Less than one hour of helicopter search time for the complete system, transportable by road and operated by two people.

Institutional. A thrust stand, certified scale, ground station, tested codebase and trained students remain with the department.

Strategic. Indian suppliers wherever they meet the specification, for sovereignty and because domestic lead times are about half those of imports.

6. Resources needed

The largest phase is INR 29,440, the first INR 26,196.

The sequence follows engineering dependency: the motor is measured before eleven more are bought, and one aircraft flies a complete mission before either of the others is begun.

Phase schedule

#	Objective	INR	Released on
1	Establish thrust and mass measurement capability	26,196	Approval
2	Verify static thrust against the 3.18 kgf requirement	5,175	Instruments commissioned
3	Aircraft 1 — autopilot and control link	25,990	**Thrust verified**
4	Aircraft 1 — onboard compute and inference	21,850	Autopilot bench-tested
5	Aircraft 1 — centimetre positioning	27,892	Compute stack operating
6	Aircraft 1 — sensing and communications	29,440	Position fix acquired
7	Aircraft 1 — airframe fabrication	20,144	Avionics integrated
8	Aircraft 1 — battery pack assembly	29,299	Airframe fabricated
9	Aircraft 1 — drive train and payload mechanism	27,098	Pack bench-discharged
10	Aircraft 1 — propulsion completed	15,525	Drive train installed
11	Charging and manual override for flight test	25,300	Aircraft assembled and weighed
12	Field equipment	17,274	Charging verified
13	Aircraft 2 — autopilot and control link	25,990	**Aircraft 1 flew a full mission**
14	Aircraft 2 — onboard compute and inference	21,850	Autopilot bench-tested
15	Aircraft 2 — centimetre positioning	27,892	Compute stack operating
16	Aircraft 2 — sensing and communications	29,440	Position fix acquired
17	Aircraft 2 — airframe fabrication	20,144	Avionics integrated
18	Aircraft 2 — battery pack assembly	29,299	Airframe fabricated
19	Aircraft 2 — drive train and payload mechanism	27,098	Pack bench-discharged
20	Aircraft 2 — propulsion completed	20,700	Drive train installed
21	Aircraft 3 — autopilot and control link	25,990	**Two aircraft, separation verified**
22	Aircraft 3 — onboard compute and inference	21,850	Autopilot bench-tested
23	Aircraft 3 — centimetre positioning	27,892	Compute stack operating
24	Aircraft 3 — sensing and communications	29,440	Position fix acquired
25	Aircraft 3 — airframe fabrication	20,144	Avionics integrated
26	Aircraft 3 — battery pack assembly	29,299	Airframe fabricated
27	Aircraft 3 — drive train and payload mechanism	27,098	Pack bench-discharged
28	Aircraft 3 — propulsion completed	20,700	Drive train installed
29	Correction source for centimetre positioning	27,892	First flight completed
30	Stable, repeatable base mounting	4,600	Base receiver acquired
31	Deployable ground segment	28,750	Base station operating
32	Statutory registration	5,750	Ground segment complete
	Total	**7,43,004**	

Phases are rounded to the rupee, so the column sums to within a few rupees of the total.

Phase 3 commits the first significant expenditure and is released only on a measured thrust figure. Phase 13 begins the second aircraft only after the first has flown a complete mission, so the largest blocks of expenditure fund replication of a demonstrated configuration rather than a projection.

Component selection

Measurement instruments — phases 1–2 INR 24,500

Item	Model	INR	Rationale
Calibrated scale	Bench scale, 30 kg, 1 g resolution, certified	12,000	Mass is a hard limit; 1 g resolves 0.02 % of the aircraft, so build-up is tracked per component. Certified, so the figure is defensible.
Thrust stand	20 kg load cell, HX711 amplifier, power meter, frame in-house	8,000	The motors ship without a thrust curve. Measuring thrust and current together gives thrust per watt. Commercial stands start near INR 45,000.
Motor (one)	Tarot TL96020, 5008, 340 KV	4,500	One measured before eleven are committed. Requirement 3.18 kgf, hover at half throttle where efficiency peaks.

Per aircraft, avionics and sensing — 4 phases INR 85,200

Item	Model	INR	Rationale
Flight controller	Holybro Pixhawk 6C Mini	22,600	Two isolated inertial units; runs ArduPilot, which our simulation is built on. F4 boards at INR 6,000 lack memory for the logging and failsafe set.
GNSS receiver	RTK-capable, RTCM3, $\leq$3 cm CEP	18,000	The largest single error term: 3.9 m on standard GNSS against 0.75 m with RTK.
Camera and lens	Arducam IMX477, 6 mm CS, fixed focus	11,100	1.03 cm/px at 40 m, or 39 pixels on a person in water. Hyperfocal 4.15 m, so fixed focus is sharp above 30 m; a focus motor can drift unnoticed.
AI accelerator	Pi AI HAT+ 26 TOPS (Hailo-8)	11,000	130–160 FPS measured against a 96 requirement. The 13 TOPS variant saves INR 4,000 but reaches only 60–80.
Radio and storage	ExpressLRS 2.4 GHz, firmware 3.5+	8,500	Native mode carries control and telemetry on one port, removing a INR 19,000 second radio. Earlier firmware consumes the link.
Onboard computer	Raspberry Pi 5, 8 GB	8,000	Hosts planner, routing, mesh and delivery logic. The only board in class exposing PCIe, which the accelerator requires.
Mesh radio	5.8 GHz node and antennas	6,000	Three video streams. 2.5 Mbps offered on a 20 Mbps channel; the constraint is link margin, not bandwidth.

Per aircraft, airframe and drive — 4 phases INR 74,100

Item	Model	INR	Rationale
Motors	Tarot TL96020, 340 KV, 4 off	18,000	3.18 kgf each, giving twice the aircraft weight in thrust. 340 KV suits an 18 in propeller at this pack voltage.
Airframe	25 × 23 mm carbon tube, machined clamps, in-house	13,000	Bending stress is 4 % of capability. Bending is not the failure mode; clamp design and vibration fatigue are, so joints are machined.
Battery cells	Molicel P45B 21700, 4500 mAh, 18 off	12,600	6S3P, 292 Wh: the mission plus a full second search plus four minutes loiter. Peak 38.3 A per cell, so a 40 A cell sits at 96 % of rating.
Pack assembly	6S 60 A BMS, distribution board, regulator	8,500	Cells arrive loose; covers welding, fusing, balancing and monitoring.
Speed controllers	50–60 A continuous, 6S, 4 off	7,200	Peak 29 A per motor, so these run near half rating. Undersizing here is a known cause of in-flight failure.
Propellers	Tarot 1855 carbon, CW/CCW, 4 off	6,500	18 in carbon at INR 3,250 per pair. The most frequently replaced item in a flight campaign.
Payload release	Four metal-gear servos, mechanical detents	4,500	Detents rather than holding torque, so a power interruption cannot release a kit.
Wiring	10 AWG silicone, XT90-S, JST-GH	2,400	Gauge set by the 115 A peak; anti-spark prevents contact pitting on a 292 Wh pack.
Propeller adapters	6 mm self-tightening hubs, 4 off	1,400	Rotation seats the nut rather than loosening it; a released propeller loses the aircraft.

Flight-test support — phases 11–12 INR 36,500

Item	Model	INR	Rationale
Battery chargers	ToolkitRC M6D, 500 W, 25 A, dual, 2 off	11,672	A 13.5 Ah pack needs about 340 W to charge in an hour. The usual HOTA D6 Pro (INR 10,499–INR 13,500) gives 325 W per channel and would run at its limit; the M6D gives 500 W at INR 5,836 and needs only a supply the department holds. Four channels charge the fleet at once.
Consumables	Fasteners, thread-lock, tape, filament	13,000	Consumed continuously in assembly and repair.
Safety-pilot radio	RadioMaster Boxer, ExpressLRS	6,955	Manual override, required every flight. Full-size gimbals: taking control of a 6.4 kg aircraft under fault conditions demands precise input.
Sun hood and monitor	Fabricated hood, second-hand monitor	1,500	Laptop displays are illegible in direct sun.

Ground segment and statutory — phases 29–32 INR 52,000

Item	Model	INR	Rationale
Field consumables	Cables, mounts, spares, transport protection	25,000	Corrections travel over the existing telemetry link, so no separate radio is needed.
Base station	Second GNSS receiver, airborne class	18,000	Compares the aircraft against a stationary receiver on a known point; without it positions degrade to 3.9 m.
Registration	Statutory, three aircraft	5,000	At 6.36 kg these are Small category under the Drone Rules 2021.
Survey tripod	Photographic tripod and adapter	4,000	A stable, repeatable mount; the corrections are relative, so survey-grade accuracy is unnecessary.

Structural cost of this arrangement

Thirty-two approvals is a material administrative overhead, and buying one aircraft at a time places the same imported order three times — roughly eight to twelve weeks of schedule. Consolidating the per-aircraft phases into one order is available if schedule becomes binding.

The detection evaluation needs no procurement: about a week on a departmental GPU and 30 GB of storage, against a public dataset.

7. Future work

Near term. Replace each calculated quantity with a measured one: detection accuracy, motor thrust, setup time. Evaluate near-infrared sensing — water absorbs strongly and skin does not, so a person appears bright against dark water. A camera without its infrared filter costs about INR 3,000, and a public dataset allows evaluation before purchase. Only the training data is flood-specific; the same system suits earthquake, wildfire and avalanche search.

Longer term. A capable aircraft deciding for itself within a very small power budget is a hard requirement in planetary aviation. *Ingenuity* flew 72 times on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth's density; *Dragonfly* launches in 2028 for Titan. The difficulty there is not flight but that a command takes minutes to arrive, so every decision is made on board. What transfers is what is not flood-specific: onboard survey planning, perception run to an energy budget, and safe behaviour without a link. One caution — microcontroller-class processors, often proposed for such missions, cannot resolve targets of this size, so a planetary configuration needs a different operating point.

8. Conclusion

Phases 1 and 2 total INR 31,371 and resolve the question the rest depends on: whether these motors lift this aircraft.

We would welcome your guidance on two decisions — the operating altitude, which the detection evaluation will inform, and whether to adopt the two-stage filter once measured.